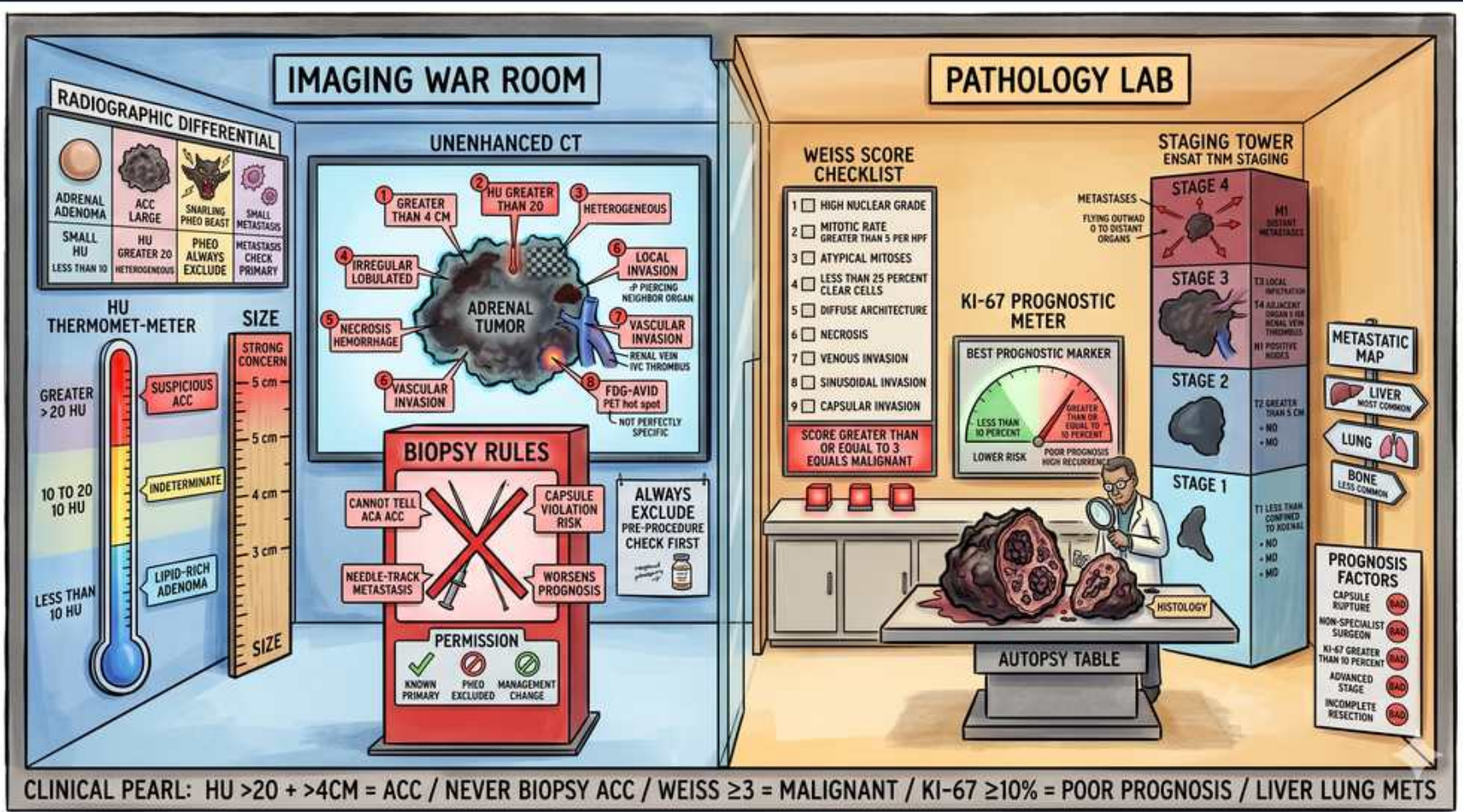


Adrenal — Imaging War Room & Pathology Lab



1. Imaging war room

WHAT YOU SEE

Blue 'IMAGING WAR ROOM' banner over the left half of the scene — the war-room framing means imaging is where the benign-vs-malignant battle over an adrenal mass is fought and decided.

WHAT IT ENCODES

Cross-sectional imaging answers the malignancy question for an adrenal mass; UNENHANCED (non-contrast) CT is the first and most important test because intracytoplasmic LIPID in benign adenomas lowers attenuation. If non-contrast CT is indeterminate, add a CT washout study or MRI chemical-shift imaging.

2. Radiographic differential panel

WHAT YOU SEE

Top-left framed thumbnails: small low-HU adenoma, large >4cm ACC, 'PHEO ALWAYS EXCLUDE'

WHAT IT ENCODES

The radiographic differential ranks lesions: small (<4 cm) low-HU homogeneous = adenoma; large (>4–6 cm), heterogeneous, irregular = ACC; macroscopic fat = myelolipoma; markedly T2-bright/'light-bulb' on MRI or very vascular = pheochromocytoma — which must ALWAYS be excluded biochemically before any intervention.

BOARD TRAP

A markedly T2-hyperintense or hypervascular mass is pheo until proven otherwise — don't catalog it as a generic adenoma and don't proceed to biopsy.

3. HU thermometer

WHAT YOU SEE

'HU THERMOMETER' — ≤10 HU lipid-rich, 10-20 indeterminate, >20 concern

WHAT IT ENCODES

Non-contrast attenuation is the single most useful number: ≤10 HU = LIPID-RICH benign adenoma (high specificity, no further workup). 10–20 HU is indeterminate (lipid-poor adenoma vs other) and needs washout/MRI; >20 HU is suspicious for malignancy/ACC, pheo, or metastasis. On CT WASHOUT, an adenoma shows absolute washout >60% or relative washout >40% even if lipid-poor.

BOARD TRAP

LOW HU is the reassuring one. The common reversal trap is treating high attenuation as benign — >20 HU raises malignancy concern, not lowers it.

4. Size ruler

WHAT YOU SEE

'SIZE' ruler with 4cm/6cm marks

WHAT IT ENCODES

Size correlates with ACC risk: <4 cm overwhelmingly benign; >4 cm raises concern; >6 cm carries a substantial malignancy probability and generally warrants resection regardless of function. Size plus attenuation (and growth >0.8–1 cm over follow-up) drive the surgical decision.

BOARD TRAP

Size alone is not diagnostic — combine with HU/phenotype. But a >4 cm lesion should never be dismissed without characterization, and rapid interval growth is a red flag.

5. >4cm + >20 HU = ACC

WHAT YOU SEE

Adrenal tumor center 'GREATER THAN 4 CM', 'HU GREATER THAN 20', 'HETEROGENEOUS'

WHAT IT ENCODES

The classic ACC imaging signature: >4 cm, non-contrast >20 HU (often >40), HETEROGENEOUS with central necrosis/hemorrhage and calcification, irregular margins, and delayed/poor contrast washout. Many ACCs are hormonally active (cortisol ± androgens → virilization, or mixed). Such a lesion should go to a multidisciplinary team for resection, not biopsy.

BOARD TRAP

Virilization + a large heterogeneous adrenal mass = ACC until proven otherwise — don't mislabel as a benign androgen-secreting adenoma.

6. Local/vascular invasion

WHAT YOU SEE

'LOCAL INVASION', 'VASCULAR INVASION'

WHAT IT ENCODES

Invasion of adjacent organs (kidney, liver) or tumor thrombus in the adrenal/renal vein or IVC is a definitive sign of MALIGNANCY (ACC) and upstages disease. It also dictates an OPEN operation by an experienced surgeon to avoid capsule violation and tumor spillage.

7. FDG-PET avid

WHAT YOU SEE

A glowing 'FDG-AVID PET' hot-spot callout on the tumor — the lit-up uptake literally shows the lesion devouring glucose, the metabolic signature of malignancy (ACC/mets) versus a cold benign adenoma.

WHAT IT ENCODES

On FDG-PET, malignant lesions (ACC, metastases) are FDG-avid with adrenal-to-liver SUV ratio above background, whereas benign adenomas are typically not. PET is most useful for indeterminate lesions, staging ACC, and detecting metastases; (18F-FDG for ACC/mets; 18F-FDOPA or MIBG for pheo/paraganglioma).

8. Biopsy rules — never biopsy

WHAT YOU SEE

Red 'CANNOT TELL ACC, NEEDLE-TRACK METASTASIS, CAPSULE VIOLATION RISK'

WHAT IT ENCODES

Adrenal biopsy is rarely indicated and is CONTRAINDICATED for suspected ACC or pheo: cytology cannot reliably distinguish adenoma from carcinoma (that requires the whole resected specimen and Weiss score), and needling risks NEEDLE-TRACK SEEDING and capsule violation that worsens prognosis. Biopsy is reserved for a suspected METASTASIS in a patient with known extra-adrenal cancer, and only AFTER pheo is biochemically excluded.

BOARD TRAP

Don't biopsy a suspected ACC (downstages/seeds, doesn't even diagnose). And never biopsy any adrenal mass before excluding pheo — needling can trigger a hypertensive crisis.

9. Always exclude pheo first

WHAT YOU SEE

'ALWAYS EXCLUDE PHEO, CHECK FIRST'

WHAT IT ENCODES

Before ANY procedure (biopsy or surgery) on an adrenal mass, biochemically exclude pheochromocytoma with plasma free or 24-h urinary fractionated metanephrines. An unsuspected pheo can release catecholamines during manipulation → lethal hypertensive crisis; if confirmed, pretreat with alpha-blockade (phenoxybenzamine/doxazosin) before beta-blockade.

BOARD TRAP

The number-one safety trap: manipulating/biopsying/operating on an adrenal mass without first ruling out pheo — phenoxybenzamine alpha-blockade must precede any beta-blocker.

10. Weiss score ≥ 3 = malignant

WHAT YOU SEE

'WEISS SCORE CHECKLIST' — high nuclear grade, mitoses, necrosis, invasion

WHAT IT ENCODES

The Weiss system is the HISTOLOGIC criterion for adrenocortical malignancy (assessable only on the resected tumor): 9 features including high nuclear grade, >5 mitoses/50 HPF, atypical mitoses, <25% clear cells, diffuse architecture, necrosis, and venous/sinusoidal/capsular invasion. A score ≥ 3 indicates carcinoma.

BOARD TRAP

Weiss requires the WHOLE specimen — this is exactly why biopsy/FNA cannot diagnose ACC.

11. Ki-67 prognostic

WHAT YOU SEE

'KI-67 PROGNOSTIC METER' — $\geq 10\%$ poor prognosis'

WHAT IT ENCODES

Ki-67 is a proliferation index in ACC: it guides adjuvant treatment and prognosis — roughly <10% lower risk, $\geq 10\%$ poor prognosis (some thresholds use >20% for high risk). Higher Ki-67 favors adjuvant mitotane $\pm$ chemotherapy.

12. ENSAT/TNM staging tower

WHAT YOU SEE

'STAGING TOWER ENSAT TNM STAGING' stages 1-4

WHAT IT ENCODES

ACC is staged by ENSAT/TNM (I–IV): I ≤ 5 cm confined, II >5 cm confined, III local invasion/regional nodes/vein thrombus, IV distant METASTASES. Stage drives therapy and prognosis — complete R0 resection is the only curative option; adjuvant mitotane for high-risk; advanced disease gets EDP-mitotane (etoposide, doxorubicin, cisplatin).

13. Metastatic map

WHAT YOU SEE

Right-side 'METASTATIC MAP' — liver, lung, bone silhouette

WHAT IT ENCODES

ACC metastasizes hematogenously most often to LIVER and LUNG, then bone; this defines stage IV (ENSAT IV). Conversely, the adrenal is a frequent recipient of metastases FROM lung, breast, melanoma, renal and GI primaries — so in a known-cancer patient a new adrenal lesion is more likely a metastasis than a primary ACC.

BOARD TRAP

Don't confuse direction: a solitary adrenal mass in a patient with known lung/breast cancer is usually a metastasis (biopsy-able after pheo exclusion), not a new ACC.

14. Clinical pearl

WHAT YOU SEE

'HU>20 + >4CM = ACC / NEVER BIOPSY ACC / WEISS ≥ 3 = MALIGNANT / KI-67 $\geq 10\%$ = POOR PROGNOSIS / LIVER LUNG METS'

WHAT IT ENCODES

Lock the bottom line: >20 HU + >4 cm + heterogeneous = ACC; never biopsy a suspected ACC or any mass before excluding pheo; Weiss ≥ 3 confirms malignancy on resection; Ki-67 $\geq 10\%$ portends poor prognosis; ACC mets favor liver and lung.